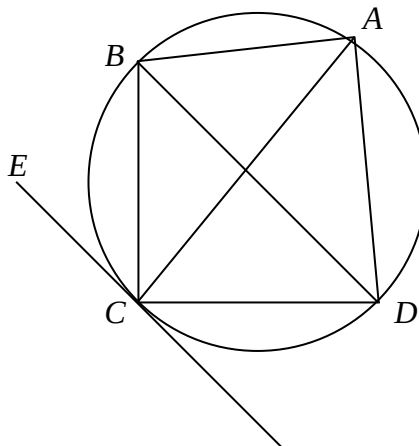


Question 1 12 Marks**Marks**

- (a) Evaluate the following definite integral $\int_{-2}^2 \frac{dx}{x^2 + 4}$ 2
- (b) Solve $\frac{5}{x-3} \geq 2$. 2
- (c) Show that $\int \frac{dx}{\sqrt{9-4x^2}} = \frac{1}{2} \sin^{-1} \frac{2x}{3} + C$, where C is constant. 2
- (d) Find the general solution for $\cos 2\theta = \frac{\sqrt{3}}{2}$ 2
- (e) (i) Show that the derivative of $\frac{1+\sin x}{\cos x}$ is $\frac{1}{1-\sin x}$. 4
- (ii) Hence, deduce that $\int_0^{\pi/4} \frac{dx}{1-\sin x} = \sqrt{2}$

Question 2 12 Marks Start a new booklet

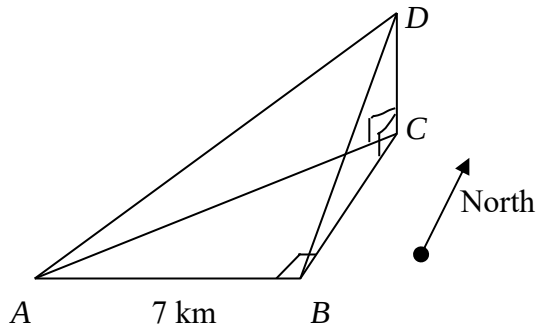
- (a) Let $f(x) = x^3 + 5x^2 + 17x - 10$. The equation $f(x) = 0$ has only one real root. 4
- (i) Show that the root lies between 0 and 2.
- (ii) Use one application of the 'halving the interval' method to find a smaller interval containing the root.
- (iii) Which end of the smaller interval found in part (ii) is closer to the root? Briefly justify your answer.
- (b) Evaluate $\int_{-1}^2 \frac{xdx}{\sqrt{3-x}}$ using the substitution $x = 3 - u$. 4
- (c) ABCD is a cyclic quadrilateral in which AC bisects $\angle DAB$. CE is the tangent to the circle at C. Prove $CE \parallel DB$. 4



Question 3 page 2.

Question 3 12 Marks Start a new booklet**Marks**

- (a) (i) Express $\sin \theta + \sqrt{3} \cos \theta$ in the form $R \sin(\theta + \phi)$. 4
- (ii) Hence, or otherwise, solve the equation $\sin \theta + \sqrt{3} \cos \theta = 1$ for values of θ between 0 and 2π .
- (b) Cadel notices that the angle of elevation of the top of a mountain due north is 14° . Upon riding 7 kilometres due west, he finds that the angle of elevation of the top of the mountain is 10° . How high is the mountain? Give your answer correct to the nearest metre. 4



$$\angle DBC = 14^\circ, \angle DAC = 10^\circ$$

- (c) (i) Show that $\cos \theta = 2 \cos^2 \frac{\theta}{2} - 1$ 4
- (ii) Hence, show that $\frac{1}{1 + \sec x} = 1 - \frac{1}{2} \sec^2 \frac{x}{2}$.
- (iii) Use part (ii) to deduce that $\int_0^{\pi/2} \frac{dx}{1 + \sec x} = \frac{\pi}{2} - 1$.

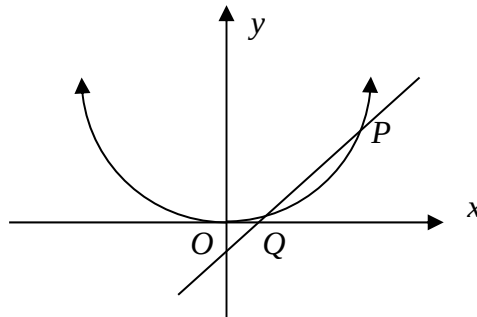
Question 4 page 3.

Question 4 12 Marks Start a new booklet

- (a) The sides of a cube are increasing at a rate of 2 cms^{-1} . Find at what rate the surface area is increasing when the sides are each 10 cm. 2
- (b) Prove by induction $9^{n+2} - 4^n$ is divisible by 5, for $n \geq 1$ 4
- (c) 100 grams of cane sugar in water is converted into dextrose at a rate which is proportional to the amount unconverted at any time. That is, if m grams are converted in t minutes, then $\frac{dm}{dt} = k(100 - m)$, where k is constant. 6
- (i) Show that $m = 100 + Ae^{-kt}$, where A is a constant, satisfies this equation.
- (ii) Find the value of A .
- (iii) If 40 grams are converted in the first 10 minutes, find how many grams are converted in the first 30 minutes.
- (iv) What is the limiting value of m as t increases indefinitely?

Question 5 12 Marks Start a new booklet

- (a) Solve the equation $6x^3 - 17x^2 - 5x + 6 = 0$, given that two of its roots have a product of -2 . 3
- (b) Find the values of a and b so that $x^4 + 4x^3 - x^2 + ax + b$ is divisible by $(x - 2)(x + 1)$. 3
- (c) $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ are points on the parabola $x^2 = 4ay$. 6



- (i) Show that the equation of the chord PQ is $\frac{(p+q)x}{2} - y = apq$
- (ii) The line PQ passes through the point $(0, -a)$. Show that $pq = 1$.
- (iv) Hence, or otherwise, if S is the focus of the parabola, show that $\frac{1}{SP} + \frac{1}{SQ} = \frac{1}{a}$.

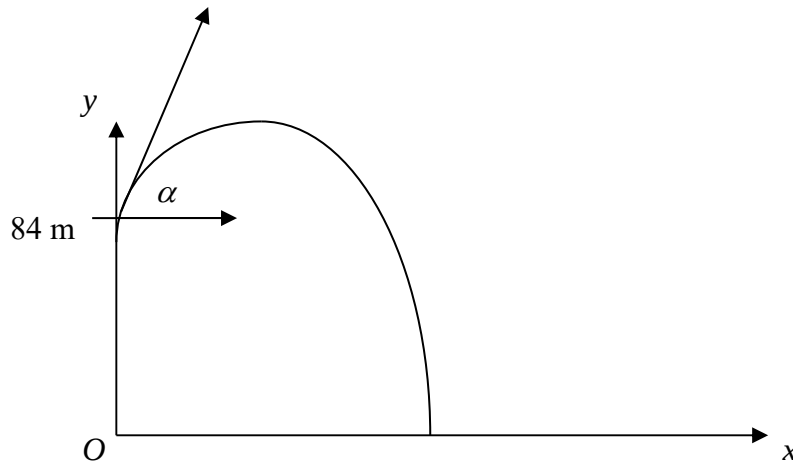
Question 6 12 Marks Start a new booklet

- (a) By considering the expansion of $(1+x)^{2n}$ in ascending powers of x show that ${}^{2n}C_0 + {}^{2n}C_1 + {}^{2n}C_2 + \dots + {}^{2n}C_{2n} = 4^n$ 2
- (b) (i) Write the expansion of $(2+3x)^{20}$ in the form $\sum_{r=0}^{20} c_r x^r$, where c_r is the coefficient of x^r in the expansion. 4
- (ii) Show that $\frac{c_{r+1}}{c_r} = \frac{60-3r}{2r+2}$
- (iii) Hence, or otherwise, find the greatest coefficient in the expansion of $(2+3x)^{20}$. Leave your answer in index form.
- (c) Of a set of otherwise similar cards, ten are white, six are red and four are yellow. Three cards are taken at random. What is the probability that: 3
- (i) They are all different colours;
- (ii) They are the same colour?
- (d) (i) In how many ways can 2 boys and 1 girl be arranged in a row if the selection is made from 4 boys and 3 girls? 3
- (ii) In how many of these arrangements does a girl occupy the middle position?

Question 7 page 5.

Question 7 12 Marks Start a new booklet

- (a) A particle is projected with speed 40 ms^{-1} from the top of a cliff 84 metres high at an angle of elevation $\alpha = \tan^{-1} \frac{4}{3}$. Assume that the equations of motion are $\ddot{x} = 0$ and $\ddot{y} = -10 \text{ ms}^{-2}$. 6



- (i) Derive the equations $x = 24t$ and $y = 84 + 32t - 5t^2$.
 - (ii) Hence or otherwise find the range on the horizontal plane through the foot of the cliff.
 - (iii) Find the speed of the body when it reaches this plane. Answer correct to two significant figures.
- (b) A particle moves in simple harmonic motion about $x = 0$ and its displacement x metres, at time t seconds, is given by $x = a \sin n(t + \alpha)$. The particle moves with a period of 16 seconds. It passes through the centre of motion when $t = 2$ seconds. Its velocity is 4 ms^{-1} when $t = 4$ seconds. 6

- (i) Show $\ddot{x} = -\frac{\pi^2}{64}x$.
- (ii) Find the maximum displacement.
- (iii) Find the speed of the particle when $t = 10$ seconds.

END OF EXAM

Question 1

$$\begin{aligned}
 \text{(a)} \quad \int_{-2}^2 \frac{dx}{x^2+4} &= 2 \int_0^2 \frac{dx}{x^2+4} \\
 &= 2 \times \left[\frac{1}{2} \tan^{-1} \frac{x}{2} \right]_0^2 \\
 &= \tan^{-1} 1 - \tan^{-1} 0 \\
 &= \frac{\pi}{4}
 \end{aligned}$$

$$\begin{aligned}
 \text{(b)} \quad \frac{5}{x-3} &\geq 2, \quad x \neq 3 \\
 \frac{5}{x-3} \times (x-3)^2 &\geq 2 \times (x-3)^2 \\
 5(x-3) &\geq 2(x-3)^2 \\
 0 &\geq 2(x-3)^2 - 5(x-3) \\
 0 &\geq (x-3)[2(x-3)-5] \\
 0 &\geq (x-3)(2x-11) \\
 3 < x &\leq \frac{11}{2} \quad \text{Note: } x \neq 3
 \end{aligned}$$

$$\begin{aligned}
 \text{(c)} \quad \int \frac{dx}{\sqrt{9-4x^2}} &= \int \frac{dx}{\sqrt{4\left(\frac{9}{4}-x^2\right)}} \\
 &= \frac{1}{2} \int \frac{dx}{\sqrt{\left(\frac{9}{4}-x^2\right)}} \\
 &= \frac{1}{2} \int \frac{dx}{\sqrt{\left(\left(\frac{3}{2}\right)^2-x^2\right)}} \\
 &= \frac{1}{2} \sin^{-1} \left(\frac{x}{\frac{3}{2}} \right) + C \\
 &= \frac{1}{2} \sin^{-1} \left(\frac{2x}{3} \right) + C
 \end{aligned}$$

$$\begin{aligned}
 \text{(d)} \quad \cos 2\theta &= \cos \frac{\pi}{6} \\
 2\theta &= 2n\pi \pm \frac{\pi}{6}, \quad n \in \{\text{integers}\} \\
 \theta &= n\pi \pm \frac{\pi}{12}
 \end{aligned}$$

(e) (i) $\frac{d}{dx} \left(\frac{1 + \sin x}{\cos x} \right) = \frac{vu' - uv'}{v^2}$, where $u = 1 + \sin x$, $v = \cos x$

$$= \frac{\cos x \cdot \cos x - (1 + \sin x) \cdot -\sin x}{\cos^2 x}$$

$$= \frac{\cos^2 x + \sin x + \sin^2 x}{\cos^2 x}$$

$$= \frac{1 + \sin x}{\cos^2 x}$$

$$= \frac{1 + \sin x}{1 - \sin^2 x}$$

$$= \frac{(1 + \sin x)}{(1 - \sin x)(1 + \sin x)}$$

$$= \frac{1}{(1 - \sin x)}$$

Note: $\cos^2 x + \sin^2 x = 1$

(ii) $\int_0^{\pi/4} \frac{dx}{1 - \sin x} = \left[\frac{1 + \sin x}{\cos x} \right]_0^{\pi/4}$

$$= \left[\frac{1 + \sin \frac{\pi}{4}}{\cos \frac{\pi}{4}} - \frac{1 + \sin 0}{\cos 0} \right]$$

$$= \frac{1 + \frac{\sqrt{2}}{2}}{\frac{\sqrt{2}}{2}} - \frac{1}{1}$$

$$= \frac{2}{\sqrt{2}} \left(1 + \frac{\sqrt{2}}{2} \right) - 1$$

$$= \frac{2}{\sqrt{2}} + 1 - 1$$

$$= \frac{2}{\sqrt{2}} = \sqrt{2}$$

Question 2

(a) (i) $f(0) = -10$, $f(2) = 52$. $\therefore f(\alpha) = 0$, $0 < \alpha < 2$

(ii) $f(1) = 13$, $\therefore 0 < \alpha < 1$

(iii) $f\left(\frac{1}{2}\right) = -\frac{1}{8}$, $\therefore \frac{1}{2} < \alpha < 1$

α is closer to 1.

(b) $x = 3 - u, \therefore \frac{dx}{du} = -1$ or $dx = -du$

When $x = -1, u = 4$ and when $x = 2, u = 1$

$$-\int_4^1 \frac{(3-u)du}{\sqrt{u}} = \int_1^4 u^{-\frac{1}{2}}(3-u)du$$

$$= \int_1^4 \left(3u^{-\frac{1}{2}} - u^{\frac{1}{2}} \right) du$$

$$= \left[6u^{\frac{1}{2}} - \frac{2}{3}u^{\frac{3}{2}} \right]_1^4$$

$$= 6.4^{\frac{1}{2}} - \frac{2}{3}.4^{\frac{3}{2}} - \left(6.1^{\frac{1}{2}} - \frac{2}{3}.1^{\frac{3}{2}} \right)$$

$$= 12 - \frac{16}{3} - \left(6 - \frac{2}{3} \right)$$

$$= \frac{4}{3}$$

(c) Let $\alpha = \angle DAC$

$\angle BAC = \angle DAC = \alpha$ (given)

$\angle BDC = \angle BAC = \alpha$ (angles in the same segment)

$\angle ECB = \angle BDC = \alpha$ (angle in the alternate segment)

$\angle CBD = \angle DAC = \alpha$ (angles in the same segment)

$\therefore \angle ECB = \angle CBD$

$\therefore EC \parallel DB$ (equal alternate angles)

Question 3

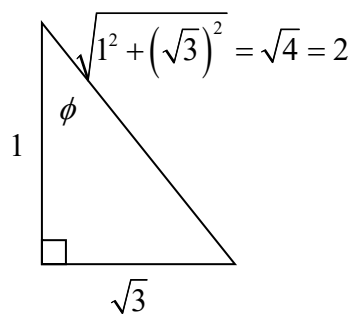
(a) (i)

$$\sin \theta + \sqrt{3} \cos \theta = 1. \sin \theta + \sqrt{3} \cos \theta$$

$$= 2 \left(\frac{1}{2} \sin \theta + \frac{\sqrt{3}}{2} \cos \theta \right)$$

$$= 2 \left(\cos \frac{\pi}{3} \sin \theta + \sin \frac{\pi}{3} \cos \theta \right)$$

$$= 2 \sin \left(\theta + \frac{\pi}{3} \right)$$



$$\sin \phi = \frac{\sqrt{3}}{2}, \cos \phi = \frac{1}{2}$$

$$\therefore \phi = \frac{\pi}{3}$$

$$\begin{aligned}
 \text{(ii)} \quad & \sin \theta + \sqrt{3} \cos \theta = 1 \\
 & 2 \sin \left(\theta + \frac{\pi}{3} \right) = 1 \\
 & \sin \left(\theta + \frac{\pi}{3} \right) = \frac{1}{2} \\
 & \theta + \frac{\pi}{3} = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{13\pi}{6}, \dots \\
 & \theta = \frac{\pi}{6} - \frac{\pi}{3}, \frac{5\pi}{6} - \frac{\pi}{3}, \frac{13\pi}{6} - \frac{\pi}{3}, \dots \\
 & \theta = -\frac{\pi}{6}, \frac{\pi}{2}, \frac{11\pi}{6}, \dots \\
 & \theta = \frac{\pi}{2}, \frac{11\pi}{6}, \text{ for } 0 \leq \theta \leq 2\pi
 \end{aligned}$$

(b)

$$\begin{aligned}
 \tan 14^\circ &= \frac{h}{BC}, \quad \therefore BC = \frac{h}{\tan 14^\circ} \\
 \tan 10^\circ &= \frac{h}{AC}, \quad \therefore AC = \frac{h}{\tan 10^\circ}
 \end{aligned}$$

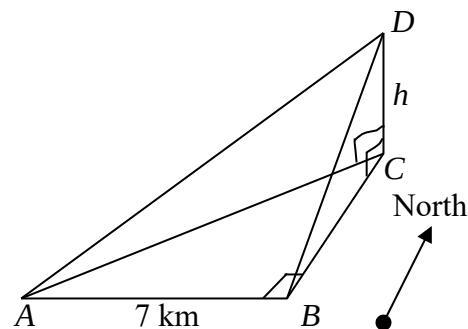
$$AC^2 = BC^2 + 7^2$$

$$\frac{h^2}{\tan^2 10^\circ} - \frac{h^2}{\tan^2 14^\circ} = 49$$

$$h^2 \left(\frac{1}{\tan^2 10^\circ} - \frac{1}{\tan^2 14^\circ} \right) = 49$$

$$h = \sqrt{\frac{49}{\frac{1}{\tan^2 10^\circ} - \frac{1}{\tan^2 14^\circ}}}$$

$$= 5.432 \text{ km (to nearest metre)}$$



$$\angle DBC = 14^\circ, \angle DAC = 10^\circ$$

$$\begin{aligned}
 \text{(c)} \quad \text{(i)} \quad & \cos \theta = \cos \left(2 \times \frac{\theta}{2} \right) \\
 & = 2 \cos^2 \frac{\theta}{2} - 1
 \end{aligned}$$

$$(ii) \quad \frac{1}{1+\sec x} = \frac{1}{1+\frac{1}{\cos x}}.$$

$$= \frac{1}{\frac{\cos x + 1}{\cos x}}$$

$$= \frac{\cos x}{\cos x + 1}$$

$$= \frac{2\cos^2 \frac{x}{2} - 1}{2\cos^2 \frac{x}{2} - 1 + 1}$$

$$= \frac{2\cos^2 \frac{x}{2}}{2\cos^2 \frac{x}{2}} - \frac{1}{2\cos^2 \frac{x}{2}}$$

$$= 1 - \frac{1}{2}\sec^2 \frac{x}{2}$$

$$(ii) \quad \int_0^{\pi/2} \left(1 - \frac{1}{2}\sec^2 \frac{x}{2}\right) dx$$

$$= \left[x - \tan \frac{x}{2} \right]_0^{\pi/2}$$

$$= \left[\frac{\pi}{2} - \tan \left(\frac{\pi}{2} \right) \right] - [0 - \tan 0]$$

$$= \frac{\pi}{2} - \tan \frac{\pi}{4}$$

$$= \frac{\pi}{2} - 1$$

Question 4

(a) Let x = length of one side of the cube. Let S = surface area = $6x^2$.

$$\frac{dx}{dt} = 2\text{cms}^{-1}, \quad \frac{dS}{dx} = 12x$$

$$\frac{dS}{dt} = \frac{dS}{dx} \cdot \frac{dx}{dt}$$

$$= 12x \cdot 2$$

$$= 24x, \text{ at } x = 10$$

$$= 240\text{cm}^2\text{s}^{-1}$$

(b) Step 1 Prove result true for $n = 1$

$$9^{1+2} - 4^1 = 9^3 - 4$$

$$= 725$$

$$= 145 \times 5$$

Step 2 Assume the result is true for $n = k$, where k is a positive integer

i.e. $9^{k+2} - 4^k = 5P$, where P is a positive integer

Step 3 Prove the result is true for $n = k + 1$

i.e. Prove $9^{k+1+2} - 4^{k+1} = 5M$

$$\text{LHS} = 9 \cdot 9^{k+2} - 4 \cdot 4^{k+1}$$

$$= 9(9^{k+2} - 4^k) + 5 \cdot 4^k$$

$$= 9(5P) + 5 \cdot 4^k, \text{ from assumption}$$

$$= 5(9P - 4^k)$$

$$= 5M, M = 9P - 4^k$$

Step 4 The result has been proved true for $n = 1, n = 1 + 1 = 2, n = 2 + 1 = 3$, etc.

Hence, by the principle of Mathematical Induction, the result is true for all positive integers.

(c) (i) $m = 100 + Ae^{-kt}$

$$\frac{dm}{dt} = -ke^{-kt}$$

$$= -k(m - 100)$$

$$= k(100 - m)$$

(ii) When $t = 0, m = 0$.

$$\therefore 0 = 100 + Ae^0, A = -100$$

(iii) When $t = 10, m = 40$

$$\therefore 40 = 100 - 100e^{-10k}$$

$$e^{-10k} = \frac{60}{100}$$

$$-10k = \ln 0.6$$

$$k = -\frac{1}{10} \ln 0.6$$

Let $t = 30$

$$m = 100 - 100e^{-\left(-\frac{1}{10} \ln 0.6\right)30}$$

$$= 100 - 100e^{3 \ln 0.6}$$

$$= 100 - 100e^{\ln(0.6^3)}$$

$$= 100 - 100(0.6^3)$$

$$= 78.4 \text{ g}$$

$$(iv) \quad \lim_{t \rightarrow \infty} m = \lim_{t \rightarrow \infty} \left(100 - 100e^{\frac{t}{10} \ln 0.6} \right)$$

$$= 100 - 0$$

$$= 100 \text{ g}$$

Question 5

- (a) $6x^3 - 17x^2 - 5x + 6 = 0$. Let the roots be $\alpha, \frac{-2}{\alpha}, \beta$.

$$\alpha + \frac{-2}{\alpha} + \beta = \frac{17}{6} \quad \text{---} \quad \boxed{1}$$

$$\alpha \cdot \frac{-2}{\alpha} \cdot \beta = \frac{-6}{6} = -1$$
$$-2\beta = -1$$

$$\beta = \frac{1}{2} \quad \text{---} \quad \boxed{2}$$

Sub $\boxed{2}$ into $\boxed{1}$.

$$\alpha + \frac{-2}{\alpha} + \frac{1}{2} = \frac{17}{6}$$

$$6\alpha^2 + -12 + 3\alpha = 17\alpha$$

$$3\alpha^2 - 7\alpha - 6 = 0$$

$$(3\alpha + 2)(\alpha - 3) = 0$$

$$\alpha = \frac{-2}{3} \text{ or } 3$$

$$\therefore \text{roots are } \frac{-2}{3}, 3, \frac{1}{2}$$

- (b) If divisible by $(x-2)(x+1)$ then $f(2) = f(-1) = 0$

$$\therefore 2^4 + 4(2)^3 - 2^2 + 2a + b = 0 \text{ and } (-1)^4 + 4(-1)^3 - (-1)^2 - a + b = 0$$

$$2a + b = -44 \quad \text{---} \quad 1.$$

$$a - b = -4 \quad \text{---} \quad 2.$$

Solve simultaneously

$$a = -16, b = -12$$

(c)

$$\begin{aligned} \text{(i)} \quad m_{PQ} &= \frac{ap^2 - aq^2}{2ap - 2aq} \\ &= \frac{a(p^2 - q^2)}{2a(p - q)} \\ &= \frac{a(p - q)(p + q)}{2a(p - q)} \\ &= \frac{p + q}{2} \end{aligned}$$

Equation PQ:

$$\begin{aligned} y - ap^2 &= \frac{p + q}{2}(x - 2ap) \\ &= \left(\frac{p + q}{2}\right)x - \left(\frac{p + q}{2}\right)2ap \\ &= \left(\frac{p + q}{2}\right)x - ap^2 - apq \\ \therefore \left(\frac{p + q}{2}\right)x - y &= apq \end{aligned}$$

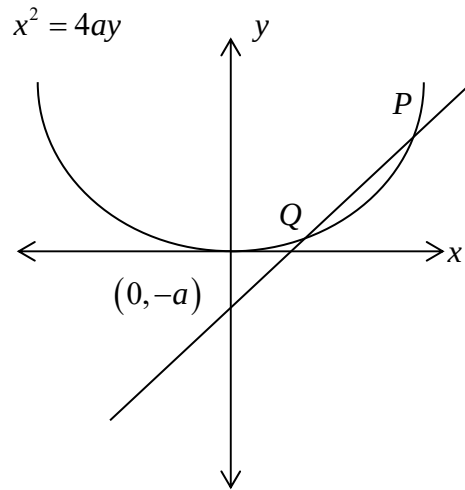
$$\text{(ii)} \quad \left(\frac{p + q}{2}\right)x - y = apq$$

Let $x = 0, y = -a$

$$0 - (-a) = apq$$

$$a = apq$$

$$pq = 1$$



$$\begin{aligned}
\text{(iii)} \quad SP &= \sqrt{(2ap-0)^2 + (ap^2-a)^2} \\
&= \sqrt{4a^2p^2 + a^2p^4 - 2a^2p^2 + a^2} \\
&= \sqrt{a^2(p^4 + 2p^2 + 1)} \\
&= \sqrt{a^2(p^2 + 1)^2} \\
&= a(p^2 + 1)
\end{aligned}$$

$$\text{Similarly } SQ = a(q^2 + 1)$$

$$\begin{aligned}
\text{Now, } \frac{1}{SP} + \frac{1}{SQ} &= \frac{1}{a(p^2 + 1)} + \frac{1}{a(q^2 + 1)} \\
&= \frac{1}{a} \left[\frac{1}{(p^2 + 1)} + \frac{1}{(q^2 + 1)} \right] \\
&= \frac{1}{a} \left[\frac{(q^2 + 1) + (p^2 + 1)}{(p^2 + 1)(q^2 + 1)} \right] \\
&= \frac{1}{a} \left[\frac{p^2 + q^2 + 2}{(p^2 + 1)(q^2 + 1)} \right] \\
&= \frac{1}{a} \left[\frac{p^2 + q^2 + 2}{p^2q^2 + p^2 + q^2 + 1} \right] \\
&= \frac{1}{a} \left[\frac{p^2 + q^2 + 2}{p^2 + q^2 + 2} \right], \quad \because pq = 1 \\
&= \frac{1}{a}
\end{aligned}$$

Question 6

$$\text{(a)} \quad (1+x)^{2n} = {}^{2n}C_0 + {}^{2n}C_1x + {}^{2n}C_2x^2 + \dots + {}^{2n}C_{2n}x^{2n}$$

$$\text{Let } x = 1$$

$$(1+1)^{2n} = {}^{2n}C_0 + {}^{2n}C_11 + {}^{2n}C_21^2 + \dots + {}^{2n}C_{2n}1^{2n}$$

$$(2)^{2n} = (2^2)^n = {}^{2n}C_0 + {}^{2n}C_1 + {}^{2n}C_2 + \dots + {}^{2n}C_{2n}$$

$$\therefore 4^n = {}^{2n}C_0 + {}^{2n}C_1 + {}^{2n}C_2 + \dots + {}^{2n}C_{2n}$$

$$\text{(b)} \quad \text{(i)} \quad (2+3x)^{20} = {}^{20}C_02^{20} + {}^{20}C_12^{19}(3x)^1 + {}^{20}C_22^{18}(3x)^2 + \dots + {}^{20}C_{20}(3x)^{20}$$

$$= \sum_{r=0}^{20} {}^{20}C_r 2^{20-r} (3x)^r$$

$$= \sum_{r=0}^{20} {}^{20}C_r 2^{20-r} 3^r x^r$$

$$= \sum_{r=0}^{20} c_r x^r, \text{ where } c_r = {}^{20}C_r 2^{20-r} 3^r$$

$$\begin{aligned}
\text{(ii)} \quad \frac{c_{r+1}}{c_r} &= \frac{{}^{20}C_{r+1} 2^{20-(r+1)} 3^{r+1}}{{}^{20}C_r 2^{20-r} 3^r} \\
&= \frac{20!}{(20-(r+1))!(r+1)!} \frac{2^{20-r-1} 3^{r+1}}{2^{20-r} 3^r} \\
&= \frac{20!}{(19-r)!(r+1)!} \times \frac{(20-r)! r!}{20!} \times \frac{3}{2} \\
&= \frac{(20-r)3}{(r+1)2} \\
&= \frac{60-3r}{2r+2}
\end{aligned}$$

$$\text{(iii)} \quad \text{Let } \frac{c_{r+1}}{c_r} > 1$$

$$\therefore \frac{60-3r}{2r+2} > 1$$

$$60-3r > 2r+2$$

$$5r < 58$$

$$r < 11.6$$

$$\therefore c_{r+1} > c_r \text{ when } r = 0, 1, 2, 3, 4, \dots, 11$$

$$\therefore c_{12} > c_{11} > c_{10} > c_9 \dots > c_2 > c_1 > c_0$$

$$\text{and } \therefore c_{r+1} < c_r \text{ when } r = 12, 13, 14, \dots, 19$$

$$\therefore c_{12} > c_{13} > c_{14} > c_{15} > c_{16} > c_{17} > c_{18} > c_{19}$$

$$\text{Maximum coefficient} = c_{12} = {}^{20}C_{12} 2^8 3^{12}$$

$$\text{(c)} \quad \text{(i)} \quad \text{Number of possible selections} = {}^{20}C_3$$

Number of ways of selecting one of each colour

$$= {}^{10}C_1 \cdot {}^6C_1 \cdot {}^4C_1$$

$$P(\text{different colours}) = \frac{{}^{10}C_1 \cdot {}^6C_1 \cdot {}^4C_1}{{}^{20}C_3} = \frac{4}{19}$$

$$\text{(ii)} \quad P(\text{same colour}) = P(3 \text{ white or } 3 \text{ red or } 3 \text{ yellow})$$

$$= \frac{{}^{10}C_3 + {}^6C_3 + {}^4C_3}{{}^{20}C_3}$$

$$= \frac{12}{95}$$

$$\text{(d)} \quad \text{(i)} \quad {}^4C_2 \cdot {}^3C_1 \cdot 3! = 108$$

$$\text{(ii)} \quad \text{Girl occupies centre position in one third of arrangements}$$

$$= \frac{1}{3} \times 108$$

$$= 36$$

Question 7

(a) (i)

Horizontal:

$$\ddot{x} = 0$$

$$\dot{x} = c$$

$$\text{When } t = 0, \dot{x} = 40 \cos \alpha$$

$$\therefore c = 40 \cos \alpha$$

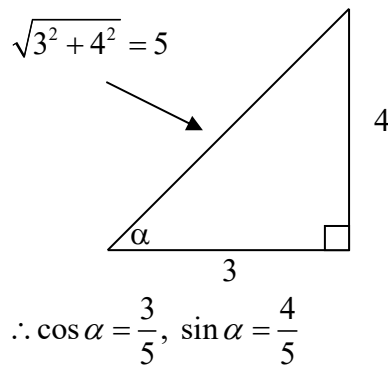
$$\therefore \dot{x} = 40 \times \frac{3}{5}$$

$$\therefore \dot{x} = 24$$

$$x = 24t + c_1$$

$$\text{When } t = 0, x = 0, \therefore c_1 = 0$$

$$\therefore x = 24t$$



Vertical

$$\ddot{y} = -10$$

$$\dot{y} = -10t + k$$

$$\text{When } t = 0, \dot{y} = 40 \sin \alpha$$

$$\therefore k = 40 \sin \alpha$$

$$= 40 \times \frac{4}{5} = 32$$

$$\therefore \dot{y} = 32 - 10t$$

$$y = 32t - 5t^2 + k_1$$

$$\text{When } t = 0, y = 84, \therefore k_1 = 84$$

$$\therefore y = 32t - 5t^2 + 84$$

(ii) Let $y = 0$

$$40t - \frac{4}{5}5t^2 + 84 = 0$$

$$32t - 5t^2 + 84 = 0$$

$$5t^2 - 32t - 84 = 0$$

$$(5t - 42)(t + 2) = 0$$

$$t = \frac{42}{5} \text{ or } -2, \text{ but } t > 0$$

$$\therefore t = 8.4 \text{ seconds}$$

Now, $x = 24t$ and the time of flight, $t = 8.4$.

Range in horizontal plane $= 24 \times 8.4 = 201.6 \text{ m}$

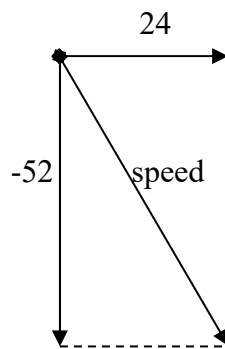
$$(iii) \quad \dot{x} = 24 \text{ ms}^{-1}$$

$$\dot{y} = 32 - 10t$$

$$= -52 \text{ ms}^{-1}$$

$$\text{Speed} = \sqrt{24^2 + (-52)^2}$$

$$= 57 \text{ ms}^{-1} \text{ (two sig. fig.)}$$



$$(b) \quad (i) \quad \frac{2\pi}{n} = 16, \quad \therefore n = \frac{\pi}{8}$$

$$\therefore x = a \sin \frac{\pi}{8}(t + \alpha)$$

$$\therefore \dot{x} = \frac{\pi}{8} a \cos \frac{\pi}{8}(t + \alpha)$$

$$\therefore \ddot{x} = -\left(\frac{\pi}{8}\right)^2 a \sin \frac{\pi}{8}(t + \alpha)$$

$$= -\left(\frac{\pi}{8}\right)^2 x$$

$$= -\frac{\pi^2}{64} x$$

$$(ii) \quad \text{When } t = 2, x = 0 \text{ and when } t = 4, \dot{x} = 4$$

$$0 = a \sin \frac{\pi}{8}(2 + \alpha)$$

$$\therefore \frac{\pi}{8}(2 + \alpha) = 0$$

$$\therefore \alpha = -2$$

$$4 = \frac{\pi}{8} a \cos \frac{\pi}{8}(4 - 2)$$

$$4 = \frac{\pi}{8} a \cos \frac{\pi}{4}$$

$$4 = \frac{\pi a}{8\sqrt{2}}$$

$$a = \frac{32\sqrt{2}}{\pi}$$

$$\therefore x = \frac{32\sqrt{2}}{\pi} \sin \frac{\pi}{8}(t - 2)$$

$$\text{Maximum displacement} = \frac{32\sqrt{2}}{\pi} \text{ m}$$

$$\begin{aligned}
 \text{(iii)} \quad \dot{x} &= \frac{\pi}{8} \cdot \frac{32\sqrt{2}}{\pi} \cos \frac{\pi}{8}(t-2) \\
 &= 4\sqrt{2} \cos \frac{\pi}{8}(t-2)
 \end{aligned}$$

Now, when $t = 10$, $\dot{x} = 4\sqrt{2} \cos \pi \text{ ms}^{-1}$

$$\begin{aligned}
 \text{Speed} &= \left| \dot{x} \right| = \left| 4\sqrt{2} \cos \pi \right| \text{ms}^{-1} \\
 &= \left| -4\sqrt{2} \right| \text{ms}^{-1} \\
 &= 4\sqrt{2} \text{ms}^{-1}
 \end{aligned}$$